CSE 309: Midterm Sample Questions

Question 1 [35 points]

Give regular expressions for the following languages over the alphabet $\Sigma = \{\mathtt{a}, \mathtt{b}\}$.

(a) (5 points) All valid integers.

(b) (5 points) All valid real numbers.

(c) (5 points) All valid complex numbers.

(d) (5 points) All strings that odd number of `a`'s and even number of `b`'s.

(e) (5 points) All strings that contain `aba` as a substring followed by `b`'s in multiple of 3.

(f) (5 points) All strings that zero or more copies of `ab` or `ba`.

(g) (5 points) All strings that have an `a` at the second position and a `b` at the third last position.

Question 2 [15 points]

Prove that the following languages are not regular.

(a) (5 points) $\{\mathtt{a}^{n-2}\mathtt{b}^n\mathtt{c}^{n+2} : n \geq 2\}$ for $\Sigma = \{\mathtt{a}, \mathtt{b}, \mathtt{c}\}$.

(b) (5 points) $\{\mathtt{0}^p : p \text{ is prime}\}$ for $\Sigma = \{\mathtt{0}\}$.

(c) (5 points) Language of matching parenthesis pairs for $\Sigma = \{\mathtt{(}, \mathtt{)}\}$.

Question 3 [5 points]

Convert following context-free grammar to Chomsky Normal Form:

$$\begin{aligned} A &\rightarrow aBCDd \mid CDE \mid \epsilon \\ B &\rightarrow BBC \mid abc \mid \epsilon \\ C &\rightarrow bb \mid \epsilon \\ D &\rightarrow cde \mid \epsilon \end{aligned}$$

Question 4 [5 points]

Design a context-free grammar for the language $\{\mathtt{a}^i\mathtt{b}^{n-i} : i \geq 0, n \geq 0\}$.

Question 5 [5 points]

Design a context-free grammar for the language that allows all valid arithmetic expressions with additions, subtraction, multiplication, and division of integers.